DATE:

Determinants

2.1 Determinant by Cofactor Expansion.

Example: 1 (Finding minor & cofactors)

$$A = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$$

Minor of a_{11}

$$M_{11} = \begin{bmatrix} \cancel{3} & \cancel{1} & \cancel{-4} \\ \cancel{2} & 5 & 6 \\ \cancel{1} & 4 & 8 \end{bmatrix}$$

$$M_{11} = \begin{vmatrix} 5 & 6 \\ 4 & 8 \end{vmatrix} = 40-24$$

$M_{11} = 16$

Cofactor of a_{11}

$C_{11} = (-1)^{1+1} M_{11}$

$C_{11} = 1(16)$

$C_{11} = 16$

Minor of a_{22}

$$M_{22} = \begin{bmatrix} 3 & \cancel{1} & -4 \\ \cancel{2} & \cancel{5} & \cancel{6} \\ 1 & \cancel{4} & 8 \end{bmatrix} = \begin{vmatrix} 3 & -4 \\ 1 & 8 \end{vmatrix}$$

$M_{22} = 24+4$

$M_{22} = 28$

Cofactor of a_{22}

$C_{22} = (-1)^{2+2} M_{22}$

$C_{22} = 28$

Remark:

$$\begin{bmatrix} + & - & + & - & + & \cdots \\ - & + & - & + & - & \cdots \\ + & - & + & - & + & \cdots \\ - & + & - & + & - & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \end{bmatrix}$$

Example:2 (Cofactor Expansion of 2x2 Matrix)

$$\begin{bmatrix} + & - \\ - & + \end{bmatrix}$$

$C_{11} = M_{11} = a_{22}$ $\qquad$ $C_{12} = -M_{12} = -a_{21}$

$C_{21} = -M_{21} = -a_{12}$ $\qquad$ $C_{22} = M_{22} = a_{11}$

Example:3 (Cofactor Expansion Along First Row)

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix} \qquad \text{Find Determinant}$$

$$\det(A) = (-1)^{1+1}\,3\begin{vmatrix} -4 & 3 \\ 4 & -2 \end{vmatrix} + (-1)^{1+2}\,1\begin{vmatrix} -2 & 3 \\ 5 & -2 \end{vmatrix} + (-1)^{1+3}\,0\begin{vmatrix} -2 & -4 \\ 5 & 4 \end{vmatrix}$$

$\det(A) = (3)(8-12) + (-1)(4-15) + 0$

$\det(A) = -12 + 11 + 0$

$\det(A) = -1$ Ans.

Example: 4 (Cofactor Expansion Along the First Column)

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix}$$

$$\det(A) = (-1)^{1+1}\,3 \begin{vmatrix} -4 & 3 \\ 4 & -2 \end{vmatrix} + (-1)^{2+1}(-2) \begin{vmatrix} 1 & 0 \\ 4 & -2 \end{vmatrix} + (-1)^{3+1}(5) \begin{vmatrix} 1 & 0 \\ -4 & 3 \end{vmatrix}$$

$$\det(A) = 3(8-12) + (-2)(-2-0) + (5)(3-0)$$

$$\det(A) = -12 - 4 + 15$$

$$\det(A) = -1 \text{ Ans.}$$

Example: 5 (Smart Choice of Row of Column)

$$A = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 3 & 1 & 2 & 2 \\ 1 & 0 & -2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix}$$

Expand with 2^{nd} col.

$$\det(A) = 0 + (-1)^{2+2} \begin{vmatrix} 1 & 0 & -1 \\ 1 & -2 & 1 \\ 2 & 0 & 1 \end{vmatrix} + 0 + 0$$

$$\det(A) = \begin{vmatrix} 1 & 0 & -1 \\ 1 & -2 & 1 \\ 2 & 0 & 1 \end{vmatrix}$$

Again Expand with 2^{nd} col.

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$$\det(A) = 0 + (-1)^{2+2}(-2)\begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} + 0$$

$$\det(A) = -2(1+2)$$

$\det(A) = -6$ ANS.

Example: 6 (Determinant of Lower Triangular Matrix)

$$\begin{vmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$$

$$\det(A) = 0 + 0 + 0 + (-1)^{4+4} a_{44} \begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\det(A) = a_{44} \begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\det(A) = a_{44}\,(0 + 0 + (-1)^{3+3} a_{33} \begin{vmatrix} a_{11} & 0 \\ a_{21} & a_{22} \end{vmatrix})$$

$$\det(A) = a_{44} a_{33} (a_{11} a_{22} - 0)$$

$$\det(A) = a_{11} a_{22} a_{33} a_{44}$$

→ So we can conclude that the det of triangular matrix will be product of diagonal entries.

A Technique For 2×2 and 3×3 matrix.

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

$$\left|\begin{matrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{matrix}\right| \begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{matrix}$$

$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$
$$- a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

Example: 7 (A Techique for Evaluating 2×2 and 3×3 Determinants)

$$\begin{vmatrix} 3 & 1 \\ 4 & -2 \end{vmatrix} = \begin{vmatrix} 3 & 1 \\ 4 & -2 \end{vmatrix} = (3)(-2) - (4)(1) = -10$$

$$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{vmatrix} = \left|\begin{matrix} 1 & 2 & 3 \\ -4 & 5 & 6 \\ 7 & -8 & 9 \end{matrix}\right| \begin{matrix} 1 & 2 \\ -4 & 5 \\ 7 & -8 \end{matrix}$$

$$= (1)(5)(9) + (2)(6)(7) + (3)(-4)(-8)$$
$$- (7)(5)(3) - (-8)(6)(1) - (9)(-4)(2)$$

$$= 45 + 84 + 96 - 105 + 48 + 72$$

$$= 240 \text{ Ans}$$

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Exercise Set 2.1:

Question: 1-2

Find all minors & cofactors of matrix A

will be solved Like Example.

Question: 3-4

4-

$$A = \begin{bmatrix} 2 & 3 & -1 & 1 \\ -3 & 2 & 0 & 3 \\ 3 & -2 & 1 & 0 \\ 3 & -2 & 1 & 4 \end{bmatrix}$$

c-

M_{41} & C_{41}

$$M_{41} = \begin{bmatrix} \cancel{2} & 3 & -1 & 1 \\ \cancel{-3} & 2 & 0 & 3 \\ \cancel{3} & -2 & 1 & 0 \\ \cancel{3} & \cancel{-2} & \cancel{1} & \cancel{4} \end{bmatrix} = \begin{vmatrix} 3 & -1 & 1 \\ 2 & 0 & 3 \\ -2 & 1 & 0 \end{vmatrix}$$

$$M_{41} = (-1)^{2+1}(2)\begin{vmatrix} -1 & 1 \\ 1 & 0 \end{vmatrix} + 0 + (-1)^{2+3}3\begin{vmatrix} 3 & -1 \\ -2 & 1 \end{vmatrix}$$

$$M_{41} = (-2)(0-1) + 0 + (-3)(3-2)$$

$$M_{41} = 2 - 3$$

$$M_{41} = -1$$

C_{41}:-

$C_{41} = +(-1)^{4+1} M_{41}$

$C_{41} = (-1)(-1)$

$C_{41} = 1$ Ans.

other parts of Q:4 & question 3 is same.

Question: 5-8

Evaluate Determinant of given Matrix if matrix is invertible, use eq: to find inverse.

8-

$$\begin{bmatrix} \sqrt{2} & \sqrt{6} \\ 4 & \sqrt{3} \end{bmatrix}$$

$\det(A) = (\sqrt{2})(\sqrt{3}) - 4\sqrt{6}$

$\det(A) = \sqrt{6} - 4\sqrt{6}$

$\det(A) = -3\sqrt{6}$

$\text{adj}(A) = \begin{bmatrix} \sqrt{3} & -\sqrt{6} \\ -4 & \sqrt{2} \end{bmatrix}$

$$A^{-1} = \frac{1}{\det(A)} \text{adj } A$$

$$A^{-1} = \frac{1}{-3\sqrt{6}} \begin{bmatrix} \sqrt{3} & -\sqrt{6} \\ -4 & \sqrt{2} \end{bmatrix} = \begin{bmatrix} \frac{-\sqrt{3}}{3\sqrt{6}} & \frac{\sqrt{6}}{3\sqrt{6}} \\ \frac{4}{3\sqrt{6}} & \frac{-\sqrt{2}}{3\sqrt{6}} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{-\sqrt{3}}{3\sqrt{6}} & \frac{1}{3} \\ \frac{4}{3\sqrt{6}} & \frac{-\sqrt{2}}{3\sqrt{6}} \end{bmatrix}$$ Ans

ALL others will be same.

Question: 9-14

Use arrow technique to find determinant

9-

$$\begin{bmatrix} a-3 & 5 \\ -3 & a-2 \end{bmatrix}$$

$\det(A) = (a-3)(a-2) + 15$

$\det(A) = a^2 - 2a - 3a + 6 + 15$

$\det(A) = a^2 - 5a + 21$ Ans

14-

$$\begin{bmatrix} c & -4 & 3 \\ 2 & 1 & c^2 \\ 4 & c-1 & 2 \end{bmatrix}$$

$$\left[\begin{array}{ccc|cc} c & -4 & 3 & c & -4 \\ 2 & 1 & c^2 & 2 & 1 \\ 4 & c-1 & 2 & 4 & c-1 \end{array}\right]$$

$\det(A) = (c)(1)(2) + (-4)(c^2)(4) + (3)(2)(c-1)$

$\quad - (4)(1)(3) - (c-1)(c^2)(c) - (2)(2)(-4)$

$\det(A) = 2c - 16c^2 + 6c - 6 - 12 - c^4 + c^3 +$

$\det(A) = -c^4 + c^3 - 16c^2 + 8c - 4$ Ans

Other would be same.

Question 15-18

Find all values of α for which $\det(A)=0$

16-

$$A=\begin{bmatrix} \alpha-4 & 0 & 0 \\ 0 & \alpha & 2 \\ 0 & 3 & \alpha-1 \end{bmatrix}$$

$$\det(A)=\alpha-4\begin{vmatrix} \alpha & 2 \\ 3 & \alpha-1 \end{vmatrix}-0+0$$

$$\det(A)=(\alpha-4)\,[\alpha(\alpha-1)-6]$$
$$\det(A)=(\alpha-4)(\alpha^2-\alpha-6)$$
$$\det(A)=(\alpha-4)(\alpha^2-3\alpha+2\alpha-6)$$
$$\det(A)=(\alpha-4)(\alpha(\alpha-3)+2(\alpha-3))$$
$$\det(A)=(\alpha-4)((\alpha-3)(\alpha+2)$$

put $\det(A)=0$

$$0=(\alpha-4)(\alpha-3)(\alpha+2)$$

$\Rightarrow \alpha=4,\ \alpha=3,\ \alpha=-2$

$$\boxed{\alpha=4,3,-2}$$

18-

$$A=\begin{bmatrix} \alpha-4 & 4 & 0 \\ -1 & \alpha & 0 \\ 0 & 0 & \alpha-5 \end{bmatrix}$$

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Expand Along 3^{rd} Column:

$$\det(A) = 0 + 0 + (-1)^{3+3}(\alpha-5)\begin{vmatrix} \alpha-4 & 4 \\ -1 & \alpha \end{vmatrix}$$

$$\det(A) = (\alpha-5)[\alpha(\alpha-4)+4]$$

$$\det(A) = (\alpha-5)(\alpha^2-4\alpha+4)$$

$$\det(A) = (\alpha-5)[\alpha^2-2\alpha-2\alpha+4)$$

$$\det(A) = (\alpha-5)[\alpha(\alpha-2)-2(\alpha-2)]$$

$$\det(A) = (\alpha-5)(\alpha-2)(\alpha-2) =$$

put $\det(A)=0$

$$0 = (\alpha-5)(\alpha-2)(\alpha-2)$$

$\Rightarrow$ $\alpha = 5$, $\alpha = 2$

$$\boxed{\alpha = 5, 2}$$

15, 17 are same.

Question: 19

Evaluate determinant in Q: 13 by Cofactor Expansion:

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & -4 \end{bmatrix}$$

d-

Along Second Column.

$$\det(A) = 0 + (-1)^{2+2}(-1)\begin{vmatrix} 3 & 0 \\ 1 & -4 \end{vmatrix} + (-1)^{3+2}\, 9 \begin{vmatrix} 3 & 0 \\ 2 & 5 \end{vmatrix}$$

$\det(A) = 0 - 1(-12-0) - 9(15-0)$

$\det(A) = 12 - 135$

$\det(A) = -123$ Ans

e-

Along 3rd Row

$$\det(A) = (-1)^{3+1}(1)\begin{vmatrix} 0 & 0 \\ -1 & 5 \end{vmatrix} + (-1)^{3+2}(9)\begin{vmatrix} 3 & 0 \\ 2 & 5 \end{vmatrix} + (-1)^{3+3}(-4)\begin{vmatrix} 3 & 0 \\ 2 & -1 \end{vmatrix}$$

$\det(A) = 1(0-0) - 9(15-0) - 4(-3-0)$

$\det(A) = 0 - 135 + 12$

$\det(A) = -123$ Ans

Question: 20

Evaluate Determinant in Q12 by Cofactor expansion

$$\begin{vmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{vmatrix}$$

Same as 19 just matrix change.

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Question: 21-26 :

Evaluate det(A) by cofactor expansion along a row or column of your choice.

23-

$$A = \begin{bmatrix} 1 & k & k^2 \\ 1 & k & k^2 \\ 1 & k & k^2 \end{bmatrix}$$

Expand along 1st col:

$$\det(A) = (-1)^{1+1} 1 \begin{vmatrix} k & k^2 \\ k & k^2 \end{vmatrix} + (-1)^{2+1} 1 \begin{vmatrix} k & k^2 \\ k & k^2 \end{vmatrix} + (-1)^{3+1} 1 \begin{vmatrix} k & k^2 \\ k & k^2 \end{vmatrix}$$

$$\det(A) = 1(k^3 - k^3) - 1(k^3 - k^3) + 1(k^3 - k^3)$$

$\det(A) = 0$ Ans.

24.

$$A = \begin{bmatrix} k+1 & k-1 & 7 \\ 2 & k-3 & 4 \\ 5 & k+1 & k \end{bmatrix}$$

Along 2nd col.

$$\det(A) = (-1)^{1+2} (k-1) \begin{vmatrix} 2 & 4 \\ 5 & k \end{vmatrix} + (-1)^{2+2} (k-3) \begin{vmatrix} k+1 & 7 \\ 5 & k \end{vmatrix} + (-1)^{3+2} (k+1) \begin{vmatrix} k+1 & 7 \\ 2 & 4 \end{vmatrix}$$

$$\det(A) = (1-k)(2k-20) + (k-3)(k^2+k-35) + (-k-1)(4k+4-14)$$

$$\det(A) = [2k - 20 - 2k^2 + 20k] + [k^3 + k^2 - 35k - 3k^2 - 3k + 105]$$

$$+ [-4k^2 - 4k + 14k - 4k - 4 + 14]$$

$\det(A) = (-2k^2+22k-20) + (k^3-2k^2-38k+105)$
$+ (-4k^2+6k+10)$

$\det(A) = -2k^2+22k-20+k^3-2k^2-38k+105-4k^2+6k+10$
$\det(A) = k^3-8k^2-10k+95$ Ans

Remaining same.

Question : 27-32

Evaluate Determinant by inspection.

27-

Diagonal matrix so
$\det(A) = (1)(-1)(1)$
$\det(A) = -1$ Ans

28-

Diagonal matrix so
$\det(A) = (2)(2)(2)$
$\det(A) = 8$ Ans

29-

Lower triangular matrix so
$\det(A) = (0)(2)(3)(8)$
$\det(A) = 0$ Ans

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30-

Upper triangular matrix so

$\det(A) = (1)(2)(3)(4)$

$\det(A) = 24$ Ans

31-

Upper Triangular matrix so

$\det(A) = (1)(1)(2)(3)$

$\det(A) = 6$ Ans

32

Lower Triangular so

$\det(A) = (-3)(2)(-1)(3)$

$\det(A) = 18$ Ans

Question: 33

Show that value of determinant is independent of θ

a-

$$\begin{vmatrix} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{vmatrix}$$

$\det(A) = \sin^2\theta + \cos^2\theta$

$\det(A) = 1$ Ans

As result is constant so its independent of θ.

b-

$$\begin{vmatrix} \sin\theta & \cos\theta & 0 \\ -\cos\theta & \sin\theta & 0 \\ \sin\theta-\cos\theta & \sin\theta+\cos\theta & 1 \end{vmatrix}$$

Expand Along C_3

$$\det(A) = 0 + 0 + (-1)^{3+3}\, 1 \begin{vmatrix} \sin\theta & \cos\theta \\ -\cos\theta & \sin\theta \end{vmatrix}$$

$\det(A) = 1(\sin^2\theta + \cos^2\theta)$

$\det(A) = 1$

↳ constant so independent

Question: 34

Show that metrices

$$A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \qquad B = \begin{bmatrix} d & e \\ 0 & f \end{bmatrix}$$

commute if & only if

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$$\begin{vmatrix} b & a-c \\ e & d-f \end{vmatrix} = 0$$

compute: AB

$$AB = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \begin{bmatrix} d & e \\ 0 & f \end{bmatrix}$$

$$AB = \begin{bmatrix} ad+0 & ae+bf \\ 0+0 & 0+cf \end{bmatrix}$$

$$AB = \begin{bmatrix} ad & ae+bf \\ 0 & cf \end{bmatrix}$$

compute : BA

$$BA = \begin{bmatrix} d & e \\ 0 & f \end{bmatrix} \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$$

$$BA = \begin{bmatrix} da+0 & db+ce \\ 0+0 & 0+cf \end{bmatrix}$$

$$BA = \begin{bmatrix} da & bd+ce \\ 0 & cf \end{bmatrix}$$

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Equate

$AB = BA$

$$\begin{bmatrix} ad & ae+bf \\ 0 & cf \end{bmatrix} = \begin{bmatrix} da & bd+ce \\ 0 & cf \end{bmatrix}$$

$ae+bf = \ +bd+ce$

$ae+bf-bd-ce = 0$

$e(a-c)+b(f-d)=0$

$-b(f-d)-e(a-c)=0$

$b(d-f)-e(a-c)=0$

Express in det. Form

$$\begin{vmatrix} b & a-c \\ e & d-f \end{vmatrix} = 0$$

so it true that matrices commute if this determinant would be 0.